

# Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: NLLB | Context: Amhara Region Agricultural and Microfinance Translation Cohort

**Output Content definition:** Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Flores-200 reference translations were produced through a multi-step workflow [Q93] with translators requiring 2-5 years of experience [Q97], 18-month re-testing [Q98], and reviewer native-speaker status with typically 5+ years of experience [Q99, Q100]. For low-resource languages, qualification thresholds were relaxed to accept broader linguistics degrees or extensive commercial experience [Q103]. No Amharic-specific annotator provenance, domain expertise, or inter-annotator agreement figures are documented in the registry. The community interview study informing research priorities included only 12 African-language speakers [Q31], a majority being immigrants in US/Europe with ~1/3 tech workers [Q32], whose perspectives 'may not consummately capture the sentiments of their communities back home' [Q39]. The deployer's required validation authority — federal bureau translators with Ethiopian agricultural/financial domain expertise [elicitation A3] — is not represented in the Flores-200 annotator workforce. Web search confirms post-publication corrections exist for Hausa/Sepedi/Xitsonga/isiZulu Flores-200 splits [WEB-14] but not Amharic, leaving Amharic reference quality unaudited. Common errors documented include 'mistranslation and unnatural translation errors' from non-English language influences and 'lower levels of standardization' [Q136-Q138]. Given OC is HIGH priority, this is a substantial concern.

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q31  | "Covering a total of 36 languages, the distribution is as follows: 5 languages are spoken predominantly in North America, 8 in South America, 4 in Europe, 12 in Africa, and 7 in Asia." (p.8)                                                                                                                                                                                                                                                                                                                                                   |
| Q32  | "Although our sample has breadth in terms of race, education, and location, the majority of our participants are immigrants living in the U.S. and Europe, and about a third of them (n = 17) identify as tech workers." (p.8)                                                                                                                                                                                                                                                                                                                   |
| Q39  | "For one, as immigrants, their perspectives may not consummately capture the sentiments of their communities back home." (p.8)                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Q93  | "We created a complex, multi-step process to ensure quality. First, professional translators and reviewers aligned on language standards. Next, translators translated the full set of Flores-200 sentences, followed by automated checks. Subsequently, the group of independent reviewers reviewed the quality, and based on their assessment, we sent some translations out for post-editing. If the quality assessment indicated that the quality is above 90 percent, the language is considered ready for inclusion in Flores-200." (p.19) |
| Q97  | "Translators are required to have at least two to three years of translation experience in the relevant language pair if they have an academic degree in translation or linguistics and three to five years of translation experience if they do not have any relevant academic qualification." (p.20)                                                                                                                                                                                                                                           |
| Q98  | "Translators also undergo a translation test every 18 months to assess their translation quality." (p.20)                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Q99  | "Flores-200 reviewers are also required to be native speakers of the target language." (p.20)                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Q100 | "Reviewers typically have a translation degree, at least five years of experience working as a translator, translation review experience, and where possible are accredited by a relevant translation board." (p.20)                                                                                                                                                                                                                                                                                                                             |
| Q103 | "In these cases, we modified the qualification process to accept applications from reviewers with more general language degrees such as Linguistics or African Language Studies, or no degree provided they have had extensive commercial translation experience (e.g. >10 years)." (p.20)                                                                                                                                                                                                                                                       |
| Q136 | "Mistranslation and unnatural translation errors were still the most common errors found while assessing the quality of the human translations." (p.22)                                                                                                                                                                                                                                                                                                                                                                                          |
| Q137 | "These were mainly due to influences from other non-English languages that may be prominently used in the target communities, leading to excessive borrowings of vocabulary and grammar, literal translations due to infrequent usage of the target language in a formal, informative content style and the lower levels of standardization." (p.22)                                                                                                                                                                                             |
| Q138 | "There has also been an increasing trend in spelling inconsistencies in the human translations due to lower levels of standardization leading to inconsistent or even subjective or preferential approaches." (p.22)                                                                                                                                                                                                                                                                                                                             |

|               |                                                                                                                                                                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WEB-14</b> | Post-publication FLORES-200 corrections exist for Hausa, Sepedi, Xitsonga, isiZulu — but not Amharic, leaving Amharic reference unaudited ( <a href="https://www.emergentmind.com/topics/flores-200-benchmark-dataset">https://www.emergentmind.com/topics/flores-200-benchmark-dataset</a> ) |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Strengths

- Documented multi-step QA workflow with 90% quality threshold [Q93, Q115, Q135] and post-editing for sub-threshold languages provides a baseline quality floor.
- Native-speaker reviewer requirement [Q99] preserves a basic legitimacy criterion for translation correctness.
- Transparency about diaspora bias in the community study [Q38, Q39, Q40] allows users to factor in selection bias when interpreting community-grounded claims.

| ID          | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q38</b>  | "We acknowledge that sampling low-resource language speakers from diasporic contexts comes with its limitations." (p.8)                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Q39</b>  | "For one, as immigrants, their perspectives may not consummately capture the sentiments of their communities back home." (p.8)                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Q40</b>  | "Over-sampling tech workers may introduce another form of selection bias." (p.8)                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Q93</b>  | "We created a complex, multi-step process to ensure quality. First, professional translators and reviewers aligned on language standards. Next, translators translated the full set of Flores-200 sentences, followed by automated checks. Subsequently, the group of independent reviewers reviewed the quality, and based on their assessment, we sent some translations out for post-editing. If the quality assessment indicated that the quality is above 90 percent, the language is considered ready for inclusion in Flores-200." (p.19) |
| <b>Q99</b>  | "Flores-200 reviewers are also required to be native speakers of the target language." (p.20)                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Q115</b> | "When full translation is completed, the QA LSP team performs a final QA review and assesses a 20% sample data. Optional arbitration, rework and QA spot checks may follow if the final quality score of the translation dataset is below 90%." (p.21)                                                                                                                                                                                                                                                                                           |
| <b>Q135</b> | "The minimum acceptable standard is 90 percent." (p.22)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                               |
|-------------|-------------------------------------------------------------------|
| <b>OC-1</b> | Do ground-truth labels reflect regional stakeholder perspectives? |
| <b>OC-2</b> | Assess potential annotator-regional population disagreement       |
| <b>OC-3</b> | Review annotator demographics from documentation                  |
| <b>OC-4</b> | Consider label re-annotation by regional annotators               |
| <b>OC-5</b> | Review aggregation methods for minority perspective erasure       |
| <b>OC-6</b> | Document label issues harming convergent/external validity        |

### Checklist responses:

**OC-1:** Ground-truth labels are general-translation-quality judgments by professional translators with 2+ years' experience [Q97]. They do not reflect Ethiopian federal bureau translator perspectives on agricultural/financial document correctness — the deployer's stated authority [elicitation A3].

**OC-2:** High potential for disagreement: the deployer requires domain-trained federal bureau translators as primary authority [elicitation A3], whereas Flores-200 annotators are general translators without documented Ethiopian government-domain expertise [Q97-Q103]. Domain-specific terminology choices (e.g., loan-penalty rendering) likely differ.

**OC-3:** Annotator demographics for Amharic specifically are not documented in the verbatim quote registry. General requirements [Q97-Q103] are documented but no Amharic-specific breakdown exists. Web search [WEB-13, WEB-14] confirms Nature 2024 paper documents the general workflow but no Amharic-specific annotator details.

**OC-4:** Re-annotation by representative regional annotator pool would be required for deployment validation. No such pool is documented; ACSI/ADCSI publish no terminology guides [WEB-15, WEB-16] that could serve as gold standard.

**OC-5:** Aggregation uses majority XSTS score per sentence then averaging [Q661]; this could erase minority perspectives, but for translation quality the more relevant concern is that minority-but-correct domain-specific renderings could be averaged out.

**OC-6:** Label issues harming validity: (a) convergent validity — annotator pool does not converge on Ethiopian government-domain perspective [elicitation A3]; (b) external validity — Flores-200 quality judgments unlikely to predict acceptance by federal bureau translators or cooperative end-users with established administrative vocabulary.

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                             |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q97    | “Translators are required to have at least two to three years of translation experience in the relevant language pair if they have an academic degree in translation or linguistics and three to five years of translation experience if they do not have any relevant academic qualification.” (p.20)                                               |
| Q98    | “Translators also undergo a translation test every 18 months to assess their translation quality.” (p.20)                                                                                                                                                                                                                                            |
| Q99    | “Flores-200 reviewers are also required to be native speakers of the target language.” (p.20)                                                                                                                                                                                                                                                        |
| Q100   | “Reviewers typically have a translation degree, at least five years of experience working as a translator, translation review experience, and where possible are accredited by a relevant translation board.” (p.20)                                                                                                                                 |
| Q101   | “We note that these are stringent standards, and extensions of Flores-200 to even more low-resource languages in the future may be difficult.” (p.20)                                                                                                                                                                                                |
| Q102   | “Already for many languages, finding a reviewer that meets the criteria above is very challenging.” (p.20)                                                                                                                                                                                                                                           |
| Q103   | “In these cases, we modified the qualification process to accept applications from reviewers with more general language degrees such as Linguistics or African Language Studies, or no degree provided they have had extensive commercial translation experience (e.g. >10 years).” (p.20)                                                           |
| Q136   | “Mistranslation and unnatural translation errors were still the most common errors found while assessing the quality of the human translations.” (p.22)                                                                                                                                                                                              |
| Q137   | “These were mainly due to influences from other non-English languages that may be prominently used in the target communities, leading to excessive borrowings of vocabulary and grammar, literal translations due to infrequent usage of the target language in a formal, informative content style and the lower levels of standardization.” (p.22) |
| Q661   | “To obtain an aggregate human quality metric for each language direction in an evaluation study, we take the majority XSTS score for each sentence and average these majority scores over all evaluated sentences.” (p.74)                                                                                                                           |
| WEB-13 | Nature NLLB paper confirms general workflow but no Amharic-specific domain expertise documentation ( <a href="https://www.nature.com/articles/s41586-024-07335-x">https://www.nature.com/articles/s41586-024-07335-x</a> )                                                                                                                           |
| WEB-14 | Post-publication FLORES-200 corrections exist for Hausa, Sepedi, Xitsonga, isiZulu — but not Amharic, leaving Amharic reference unaudited ( <a href="https://www.emergentmind.com/topics/flores-200-benchmark-dataset">https://www.emergentmind.com/topics/flores-200-benchmark-dataset</a> )                                                        |
| WEB-15 | ACSI official website — no Amharic financial terminology guide publicly available ( <a href="http://www.acsi.org.et/">http://www.acsi.org.et/</a> )                                                                                                                                                                                                  |
| WEB-16 | FinDev Gateway ACSI profile — no terminology resource ( <a href="https://www.findevgateway.org/case-study/2004/01/amhara-credit-and-saving-institution-acsi-institutional-profile-current-status">https://www.findevgateway.org/case-study/2004/01/amhara-credit-and-saving-institution-acsi-institutional-profile-current-status</a> )              |

## Information Gaps

- No public record of who specifically annotated the Amharic Flores-200 split, their credentials, or their domain backgrounds.
- No published Amharic-specific Flores-200 error audit comparable to those for Hausa/Sepedi/Xitsonga/isiZulu [WEB-14].

## Requires Expert Verification

- Whether federal bureau translators would actually disagree with Flores-200 Amharic reference renderings on general-domain Wikimedia content (a separate question from domain-specific document fitness).

## Remediation

**Gap 1: Amharic Flores-200 annotator credentials and provenance not documented [no Q-evidence]; annotators are general professional translators [Q97-Q103], not Ethiopian government-domain specialists; no Amharic-specific post-publication audit exists [WEB-14].**

**Recommendation:** Commission an Amharic Flores-200 audit analogous to the Abdulmumin et al. corrections for other African languages, and recruit a panel of federal bureau translators to re-annotate a domain-specific evaluation sample, treating their judgments as the convergent-validity gold standard.

| ID     | Evidence                                                                                                                                                                                                                                                                                               |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q97    | “Translators are required to have at least two to three years of translation experience in the relevant language pair if they have an academic degree in translation or linguistics and three to five years of translation experience if they do not have any relevant academic qualification.” (p.20) |
| Q98    | “Translators also undergo a translation test every 18 months to assess their translation quality.” (p.20)                                                                                                                                                                                              |
| Q99    | “Flores-200 reviewers are also required to be native speakers of the target language.” (p.20)                                                                                                                                                                                                          |
| Q100   | “Reviewers typically have a translation degree, at least five years of experience working as a translator, translation review experience, and where possible are accredited by a relevant translation board.” (p.20)                                                                                   |
| Q101   | “We note that these are stringent standards, and extensions of Flores-200 to even more low-resource languages in the future may be difficult.” (p.20)                                                                                                                                                  |
| Q102   | “Already for many languages, finding a reviewer that meets the criteria above is very challenging.” (p.20)                                                                                                                                                                                             |
| Q103   | “In these cases, we modified the qualification process to accept applications from reviewers with more general language degrees such as Linguistics or African Language Studies, or no degree provided they have had extensive commercial translation experience (e.g. >10 years).” (p.20)             |
| WEB-14 | Post-publication FLORES-200 corrections exist for Hausa, Sepedi, Xitsonga, isiZulu — but not Amharic, leaving Amharic reference unaudited ( <a href="https://www.emergentmind.com/topics/flores-200-benchmark-dataset">https://www.emergentmind.com/topics/flores-200-benchmark-dataset</a> )          |